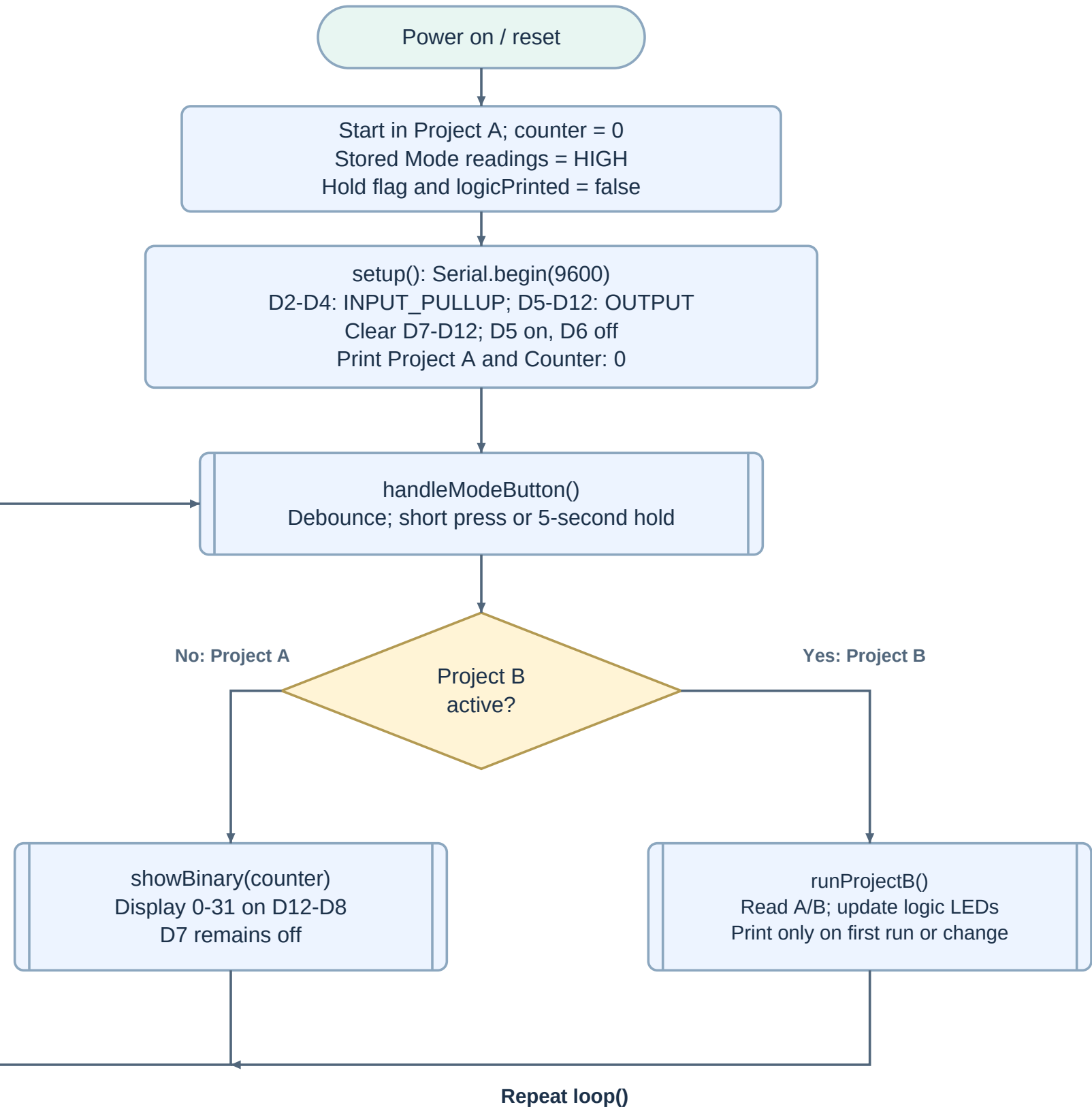


Main program flow

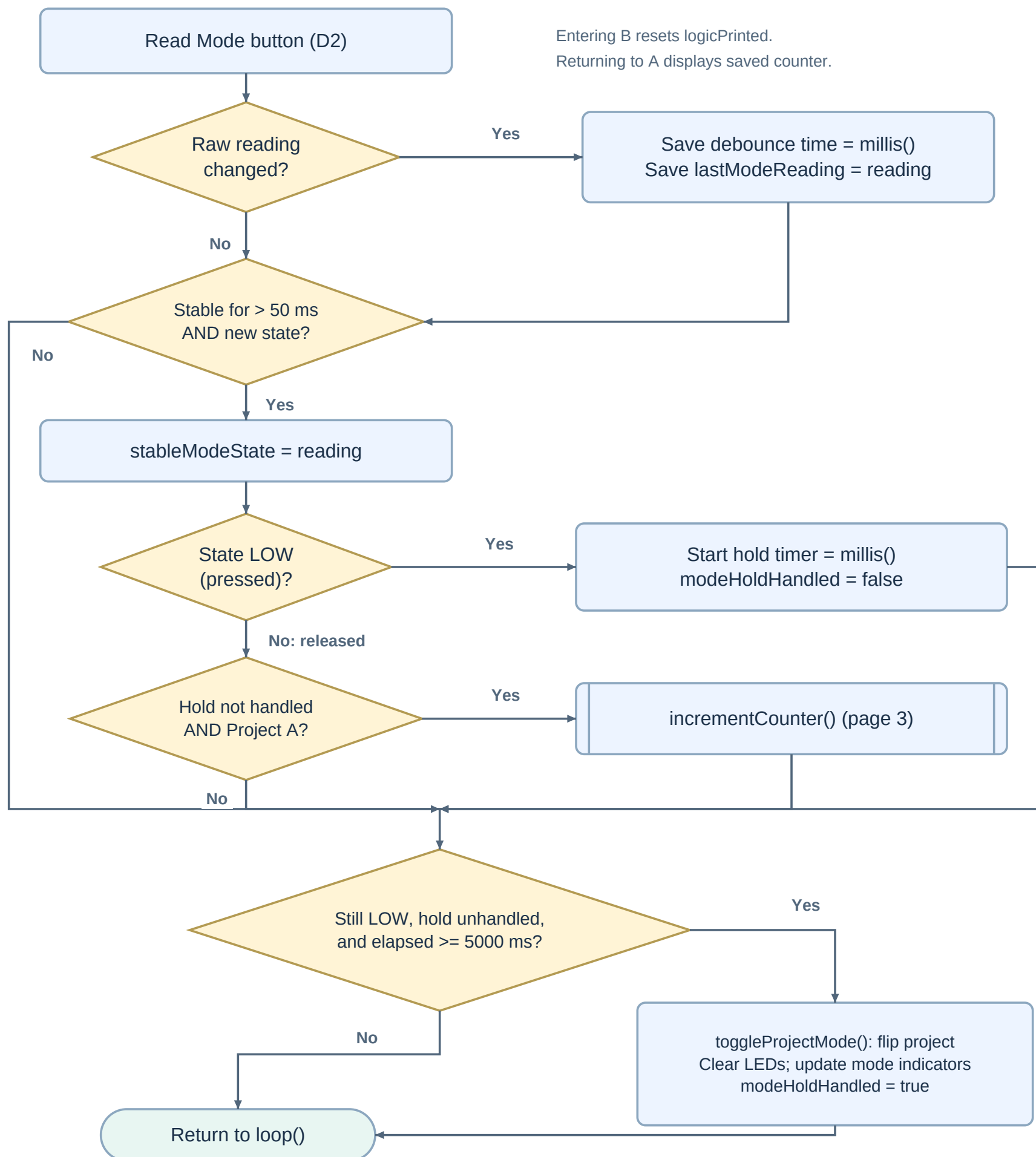
setup() runs once; loop() repeats continuously.



Mode button: D2. Logic buttons A/B: D3/D4. LOW means pressed.
Short Mode press increments on release in Project A.
Holding Mode for at least 5 seconds switches projects once per hold.
The counter value is preserved when switching modes.

Mode button handling

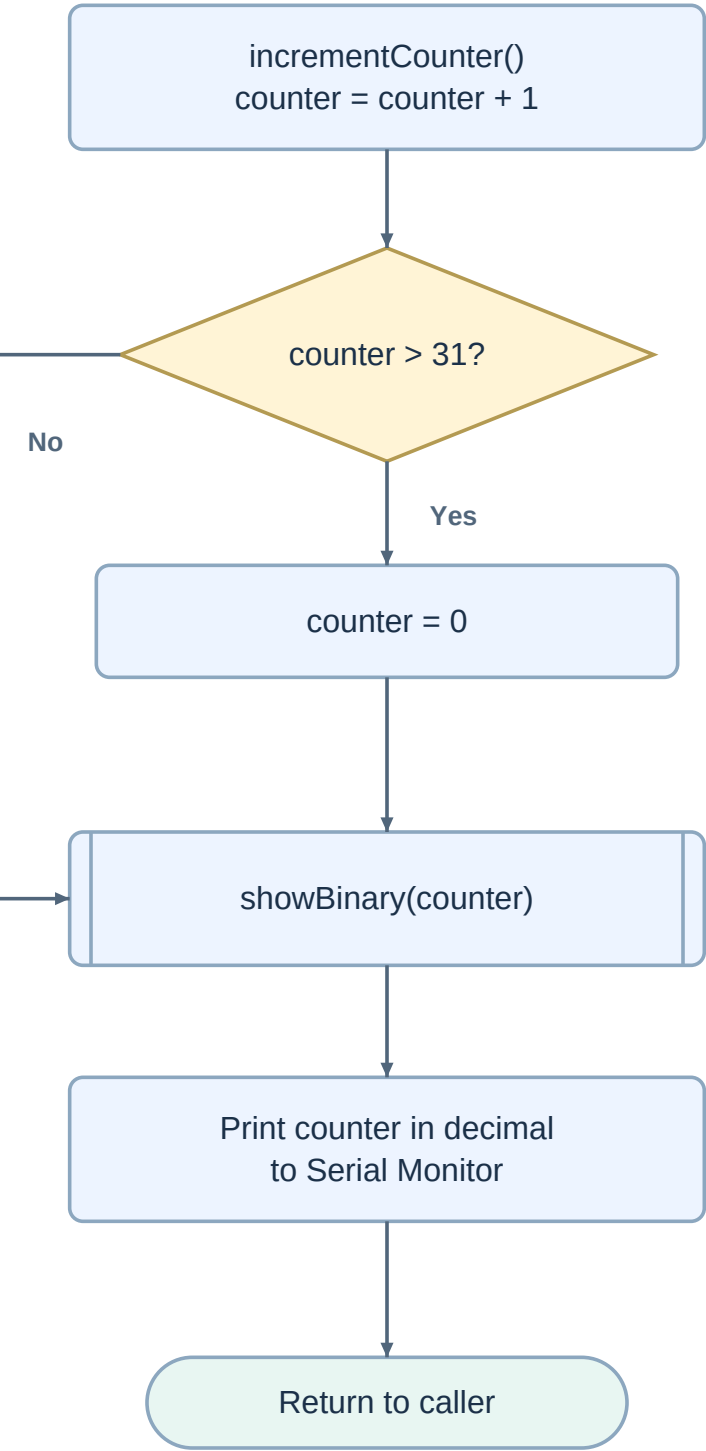
handleModeButton(): debounce first, then always check the hold timer.



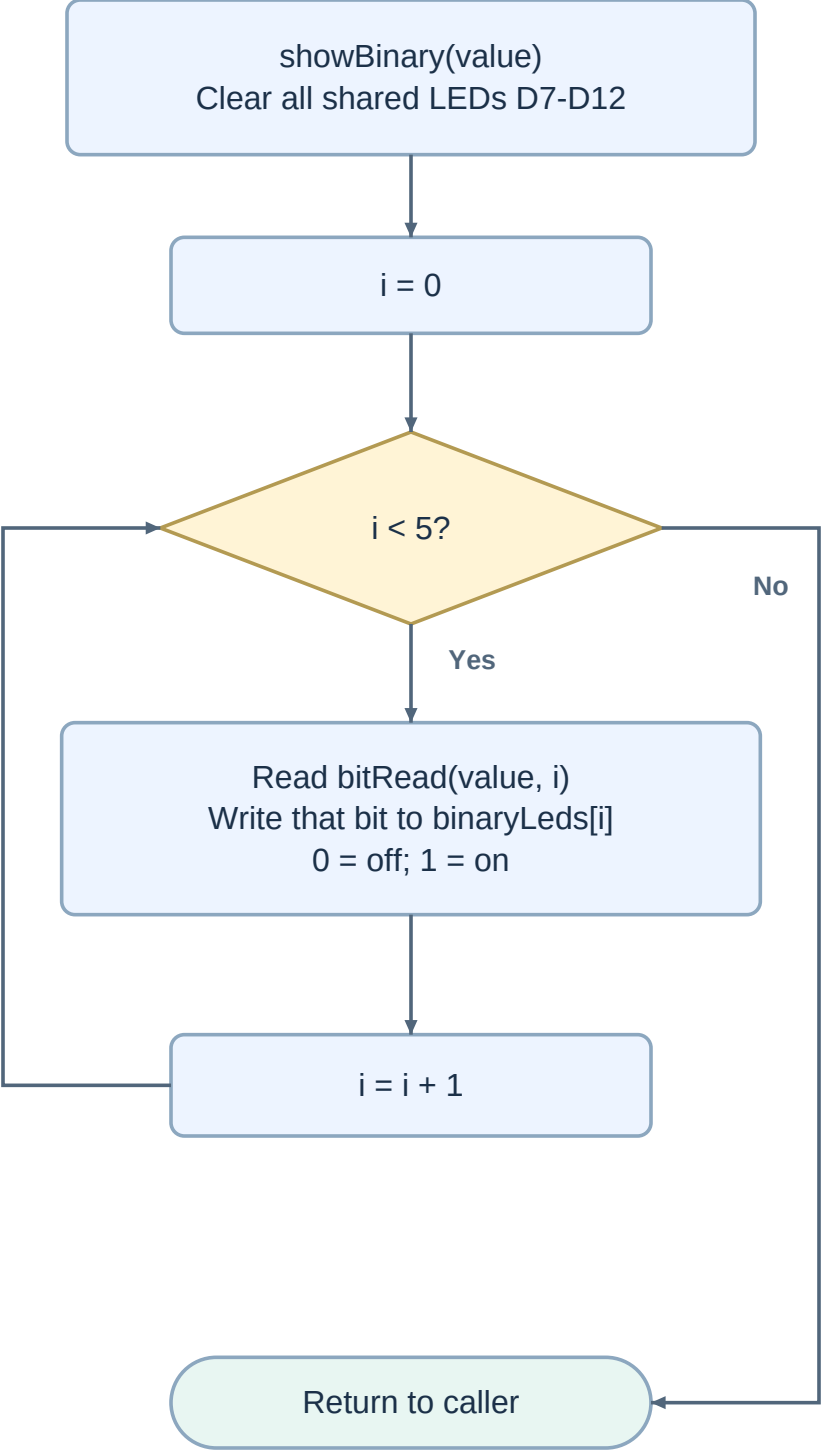
Project A: binary counter

incrementCounter() changes the value; showBinary(value) draws the five bits.

INCREMENT ON SHORT RELEASE



BINARY DISPLAY



Binary pin order: D12 = 1, D11 = 2, D10 = 4, D9 = 8, D8 = 16.
D7 stays off in Project A. Five bits represent 32 values: 0 through 31.
turnOffDataLeds() loops through all six shared pins and writes LOW.

Project B: logical operators

runProjectB() refreshes LEDs every pass, and prints only when needed.

